# PARISHRAM 2.0

## CLASS 12TH 2026

# Relations and Functions

**MATHS** **LECTURE-1**

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# TOPICS TO BE COVERED

1. What are Sets? (11th)
2. Cartesian Product ( )
3. What Are Relations
4. Domain, Range and Codomain

# What Are Sets?

Collection of well defined elements.

Well defined Statement ✓

Statement → Sabke answer same ✓

- Natural numbers less than 5

→ {1, 2, 3, 4} → elements

- Rich people in city.

Mumbai

→ Ambani
→ Mandar
→ Rohit S

Well defined ✗

Fav actor

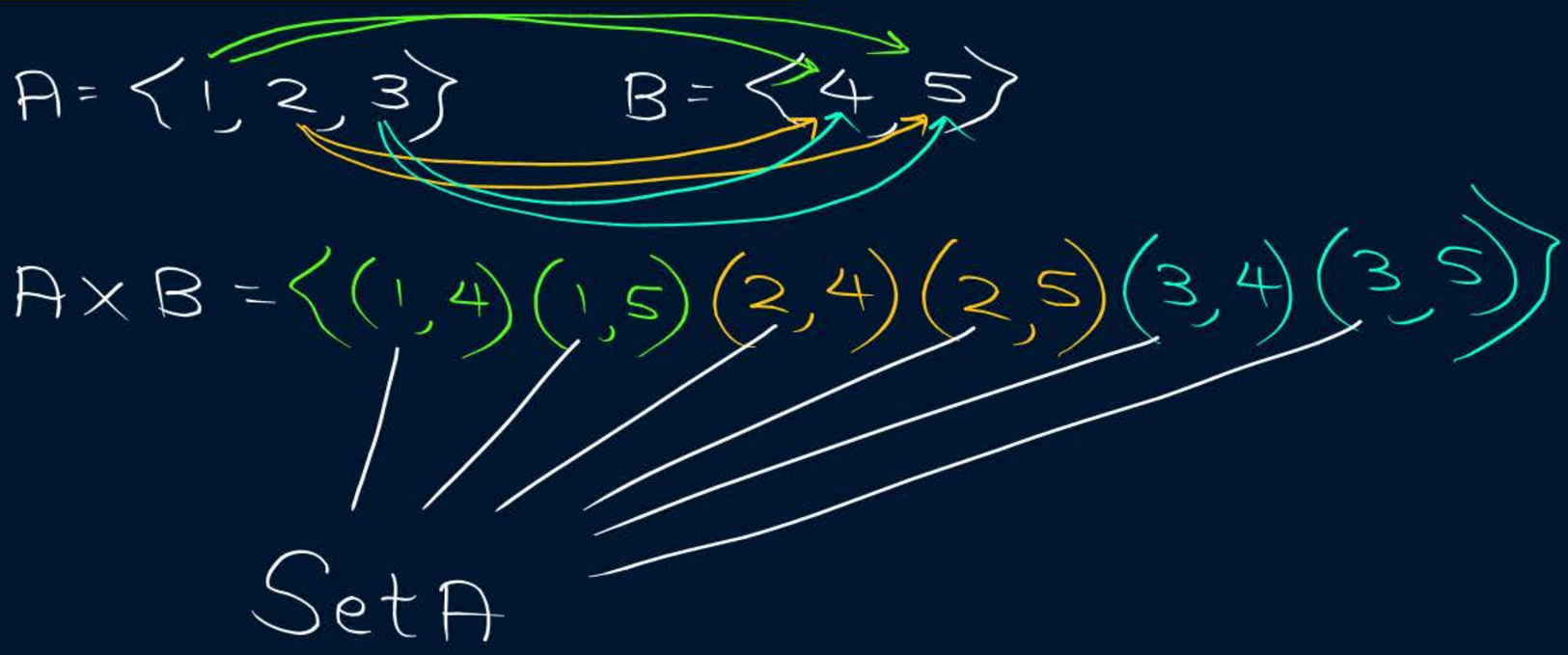
→ different answer

Well defined ✗

# Cartesian Product

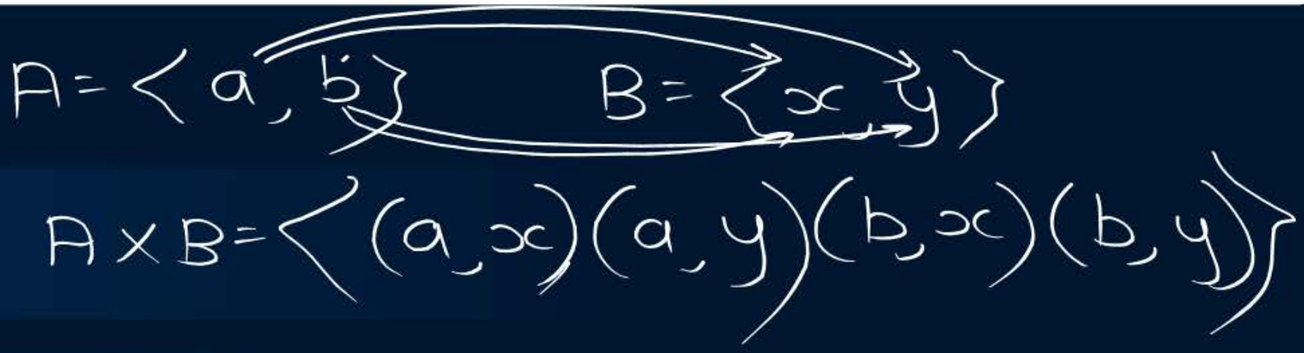
Sets Ka Multiplication

$$A \times B = \{(a, b), a \in A, b \in B\}$$



# QUESTION

$A = \{a, b\}, B = \{x, y\}$. Find $A \times B$.



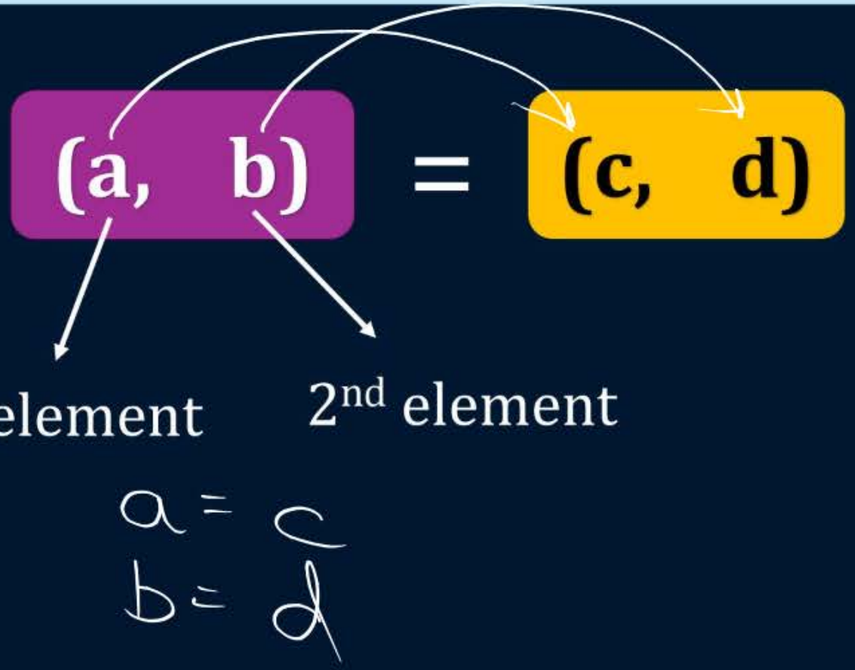

1. $(a, x), (a, y), (b, x), (b, y)$
2. $(a, x), (a, y)$
3. $(x, a), (x, b), (y, a), (y, b)$
4. None of these

# Ordered Pair → Order Matter ✓

An ordered pair is a pair of elements written as $(a, b)$ where ordered matters.

$(a, b) \neq (b, a)$ unless $a = b$.

**Ordered pair is a collection of elements.**




$(a, b) \neq (b, a)$

$(x, y) = (7, 5)$

$x = 7$

$y = 5$

# POLL QUESTION

**If $(x + 5, y - 4) = (12, 3)$. Find $x$ and $y$.**

Ordered Pair

1. (3, 7)
2. (7, 7)
3. (6, 3)
4. (4, 4)

$x + 5 = 12$

$x = 12 - 5$

$x = 7$

$y - 4 = 3$

$y = 3 + 4$

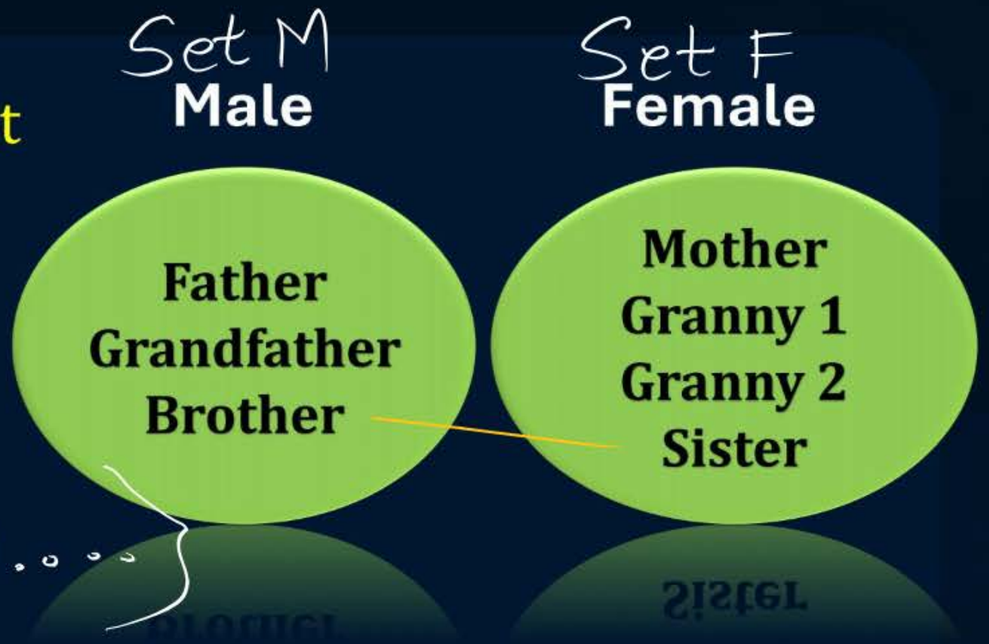
$y = 7$

# Relations

$R \subseteq A \times B$

If $A$ and $B$ are two non-empty sets, then any subset $R$ of $A \times B$ is called relation form set $A$ to $B$.

i.e. $R : \boxed{A \to B} \Leftrightarrow \boxed{R \subseteq A \times B}$




Example:

$M \times F = \{(Father, Mother), (Fath, G1)(F, G_2) \ldots\}$

Cartesian Product

$R_1 = \{$Couple from family$\}$

$R_1 = \{(Father, Mother), (Grand F, Granny 1), (Grand F, Granny 2)\}$

$R_2 = \{$Siblings from Family$\}$

$R_2 = \{(Brother, Sister)\}$

$\boxed{R_1 \subseteq M \times F}$

$\boxed{R_2 \subseteq M \times F}$

# Subset

$A = \{1, 2, 3\}$

$B = \{3, 4, 5\}$

$C = \{1, 2, 3, 4, 6, 7\}$

Set B → Saare elements → Set C ✗

B is not a subset of C

Kya Set A → Saare elements → Set C → A is a subset of C

# Relations

$A = \{1, 2, 3\}$ $B = \{4, 5\}$

$A \times B = \{(1,4)(1,5)(2,4)(2,5)(3,4)(3,5)\}$

1. $R_1 = \{(x, y) : y = x + 3, x \in A, y \in B\}$

$x = 3$, $y = 3 + 3$, $y = 6$

$R_1 = \{(1,4)(2,5)(3,6)\} = \{(1,4),(2,5)\}$

$y = x + 3$

$x = 1$, $y = 1 + 3 = 4$ $(1,4)$

2. $R_2 = \{(x, y) : y > x, x \in A, y \in B\}$

$R_2 = \{(1,4)(1,5)(2,4)(2,5)(3,4)(3,5)\}$

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**Total Number of Relations**

$R : A \to B$

**For set $A$ and set $B$**

$2^{n(A) \times n(B)}$

$2^{n(A) \times n(B)}$

$A = \{a, b, c\}$ Set A $n(A) = 3$

$B = \{1, 2, 3, 4\}$ Set B $n(B) = 4$

Total No of Relations $(A \to B) = 2^{n(A) \times n(B)}$

$= 2^{3 \times 4}$

$= 2^{12}$

# POLL QUESTION

**$A = \{1, 2, 3\}$ and $B = \{a, b\}$. Then number of relations possible?**

$n(A) = 3 \quad n(B) = 2$

$= 2^{n(A) \times n(B)}$

$= 2^{3 \times 2}$

$= 2^6$

$= 64$

1. 32
2. 64 ✓
3. 128
4. 16

# Domain, Range and Co-domain

$A = \{1, 2, 3\}$

$B = \{1, 2, 3, 4, 5, 6\}$

$R = \{(x, y), y = 2x, x \in A, y \in B\}$

$R = \{(1,2), (2,4), (3,6)\}$

Domain [Input] $= \{1, 2, 3\}$

Range [output] $= \{2, 4, 6\}$

Codomain $= \{1, 2, 3, 4, 5, 6\}$

Relation A → B → some elements

Domain

$y = 2x$

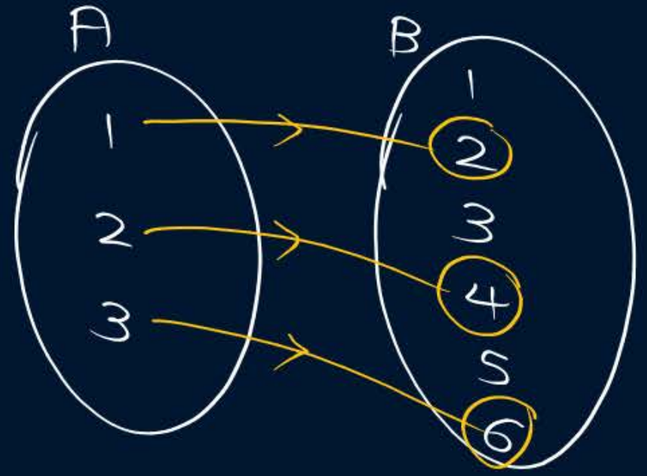
$x = 1, y = 2(1) = 2$ ✓

$x = 2, y = 2(2) = 4$ ✓

$x = 3, y = 2(3) = 6$ ✓

$x = 4$ ✗

Range

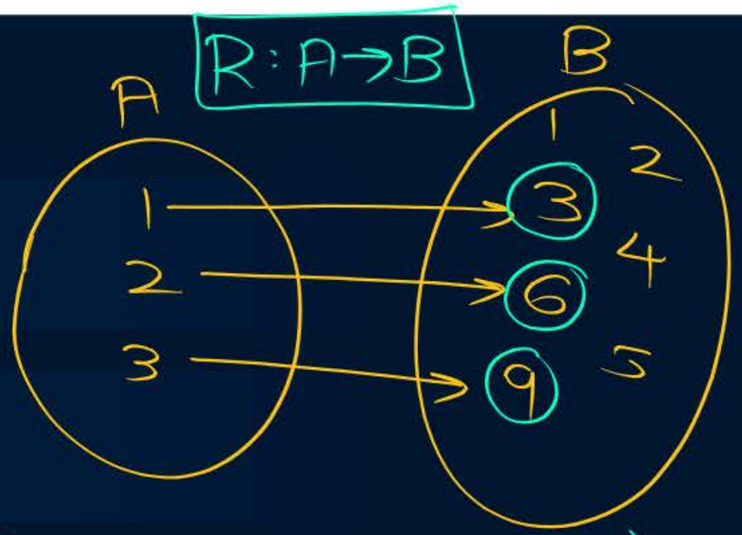


# POLL QUESTION

$R : A \to B$

**In a relation $R$ from set $A$ to set $B$, the set $B$ is called the:**

1. Domain of $R$
2. Range of $R$
3. Codomain of $R$
4. None of the above




Domain → {1, 2, 3}

Range → {3, 6, 9}

Co-domain → B (Saare elements)

↳ {1, 2, 3, 4, 5, 6, 9}

# THANK YOU